3-29-23

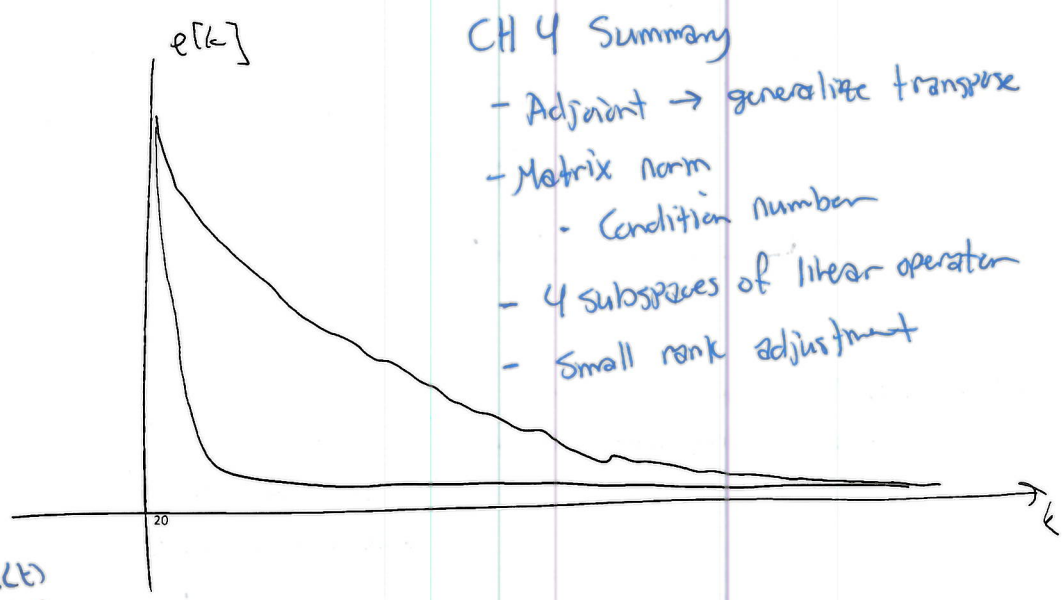
$$A = \begin{bmatrix} f[1] & \cdots & f[1-m] \\ \vdots & & \\ f[N] & \cdots & f[N-m] \end{bmatrix}$$

$$= \begin{bmatrix} \text{---} & q[1]^H & \text{---} \\ \text{---} & q[2]^H & \text{---} \\ & \vdots & \\ \text{---} & q[N]^H & \text{---} \end{bmatrix} \qquad A^H = \begin{bmatrix} | & | & & | \\ q[1] & q[2] & \cdots & q[N] \\ | & | & & | \end{bmatrix}$$

$$(A^H A) = \sum_{i=1}^{N} q[i]\, q[i]^H$$

Add one more row:

$$(A^H A) = \sum_{i=1}^{N+1} q[i]\, q[i]^H = \underbrace{\sum_{i=1}^{N} q[i]\, q[i]^H}_{\text{old } A} + \underbrace{q[N+1]\, q[N+1]^H}_{\text{a rank-one adjustment.}}$$



CH 4 Summary

- Adjoint → generalize transpose
- Matrix norm
  - Condition number
- 4 subspaces of linear operator
- Small rank adjustment

The solution of this minimizes the error of d-y by creating an H(t) that minimizes over all time 0 → T.

We want the H we find to reflect current circumstances, to do this we weight our error

$$J(n) = \sum_{i=1}^{n} \lambda^{n-i} |e(i)|^2 \qquad \lambda = 0.95$$

set it less than but close to 1

The older it is, the less it matters to us.

Expand

$$= |e(n)|^2 + \lambda |e(n-1)|^2 + \lambda^2 |e(n-2)|^2 + \dots$$

- We will derive the equation using λ. The non-λ derivation is in the book

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Chapter 5 ?

Matrix Factorizations

- LU — General matrix, square
- QR — General matrix
- Cholesky — self-adjoint, square $\left.\begin{array}{l} A^T = A \\ A^H = A \end{array}\right\} A^* = A$

I) LU

for general square matrices

$$A = LU = [\text{Lower triangular mat}][\text{upper triangular mat}]$$

$$= \left[\triangle\right]\left[\triangle\right]$$

Gaussian Elimination

The diagonal of L matrix is to ones

$$A\underline{x} = \underline{b}$$

$$\boxed{\begin{array}{l} - \; LUx = b \\ \rightarrow \; Ux = y \; (1) \\ \quad Ly = b \; (2) \end{array}}$$

$$\left[\triangle\right]\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} \qquad \begin{bmatrix} l_{11} & \\ l_{21} & l_{22} \end{bmatrix}\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$\underline{y} = L^{-1}\underline{b}$$

$$l_{11} y_1 = b_1 \longrightarrow y_1 = \frac{b_1}{l_1}$$

$$l_{21} y_1 + l_{12} y_2 = b_2$$

$$y_2 = \frac{b_2 - l_{21} y_1}{l_{12}}$$

$$\vdots$$

With this structure, we don't have to invent any matrices

Because we have cascading solutions

Backsubstitution.

Now that we solved for y lets solve for b in (2)

$Ux = y$:

$$\left[\triangle\right]\begin{bmatrix} x \end{bmatrix} = \begin{bmatrix} y \end{bmatrix}$$

$x = A \backslash b$

~~$x = \text{inv}(A) * b$~~

A) Determinant of L

$$\det(A) = \det(Lu)$$
$$= \det(L) \cdot \det(u)$$
$$= 1 \cdot \prod_{i=1}^{n} u_{ii}$$

Because L has 1 down the diagonal

$\hookrightarrow \det(L) = 1$

B)

Cholesky

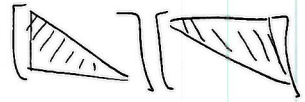
$A = A^*$ — The matrix is ~~[illegible]~~ self-adjoint

Real

$A = LL^T \longleftrightarrow A = LL^H$

Complex

Similarly to before, we can break it apart

$=$ 

$Z \sim N(0, I)$

randn

$X = LZ$

$E[X] = E[LZ]$

$= L\,E[Z] = 0.$

$\text{cov}(X) = E[XX^T] - \cancel{E[X]}^{\,0}\,E[X^T]$

$= E[LZZ^TL^T]$

$= L\,E[ZZ^T]\,L^T$

$= \underbrace{LL^T} = R$

* II) QR

We can do this with the Gram-Schmidt Process!

A: general.

$A = QR$

where: Q is $\nearrow$ orthogonal (A is real) $Q^TQ = I$

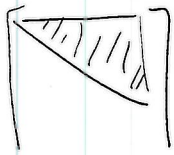
$\searrow$ unitary (A is complex) $Q^HQ = I$

$$Q = \begin{bmatrix} | & | & & | \\ q_1 & q_2 & \cdots & q_n \\ | & | & & | \end{bmatrix}$$

$$Q^T = \begin{bmatrix} - & q_1^T & - \\ - & q_2^T & - \\ & \vdots & \\ - & q_n^T & - \end{bmatrix} \qquad Q^T Q = I$$

diagonals are 1 so this is normalized

It is a "diagonal as this can be"

$R =$ 

$$\min \| A x - b \|_2^2 \qquad \hat{x} = \underbrace{(A^H A)^{-1}}_{} A^H \underline{b}$$

This causes major numerical problems

Suppose we have done QR factorization somehow

$$\| Q R \underline{x} - b \|_2^2$$

$$\| \underset{\sim}{Q} (R \underline{x} - Q^H \underline{b}) \|_2^2$$

$$= \| R \underline{x} - \underbrace{Q^H b}_{} \|_2^2$$

use this

to write this

If Q is a unitary matrix:

$$\| Q x \|_2^2 = \| x \|_2^2$$

$$\langle Qx, Qx \rangle = x^H Q^H Q x = x^H x = \| x \|_2^2$$

$$\left\| \begin{array}{c} R \\ \begin{bmatrix} R_1 \\ 0 \end{bmatrix} \end{array} \underline{x} - \begin{array}{c} c \\ \begin{bmatrix} c \\ d \end{bmatrix} \end{array} \right\|_2^2$$

$$\begin{bmatrix} R_1 \\ \underline{0} \end{bmatrix} \underline{x} - \begin{bmatrix} \underline{c} \\ \underline{d} \end{bmatrix}$$

$$\left\| \begin{bmatrix} R_1 \underline{x} - \underline{c} \\ 0 \underline{x} - \underline{d} \end{bmatrix} \right\|_2^2 \longrightarrow \text{set } \underline{x} = \underbrace{R_1^{-1} \underline{c}}_{}$$

Can use backsubstitution to solve this

$$\left\| \begin{bmatrix} \underline{0} \\ -\underline{d} \end{bmatrix} \right\|_2^2$$